

IS 6813 MSBA Capstone Project

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Business Problem

Qx conversion rate
19.7%

Mx conversion rate
12.7%

performance gap
(**≈55%** relative
difference)

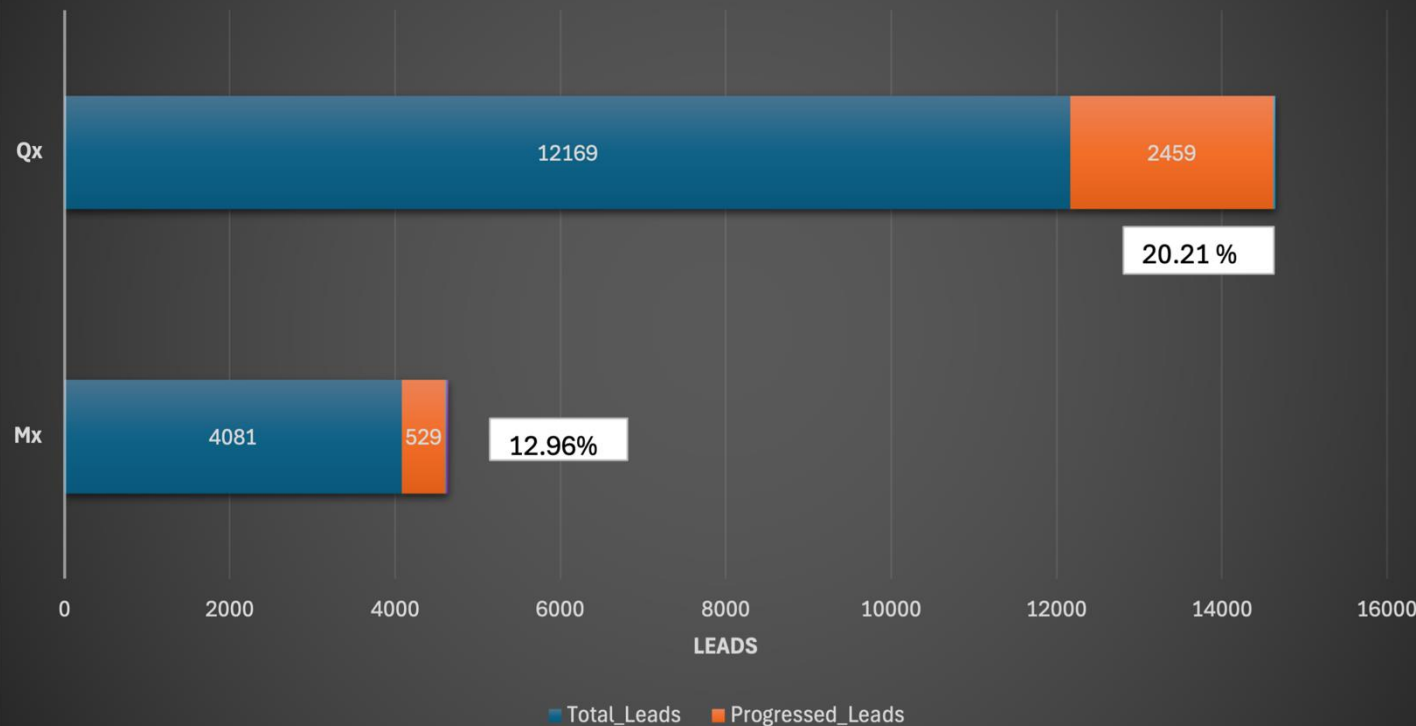
Who is the ideal Mx Customer
(ICP)?

- ✓ Industries
- ✓ Company Types

- ✓ Job Titles
- ✓ Cross-Sell Opportunities

EDA Overview: Understanding Mx Underperformance

Overall Conversion Baseline — Mx vs Qx



What drives lead progression?

Data Quality & Cleaning

Data Structure

- ✓ 16,644 Rows
- ✓ 14 Variable

Class Imbalance

- ✓ Strong class imbalance with only about 13% of Mx leads progressing
- ✓ The outcome is rare, but valuable.

Missing Values

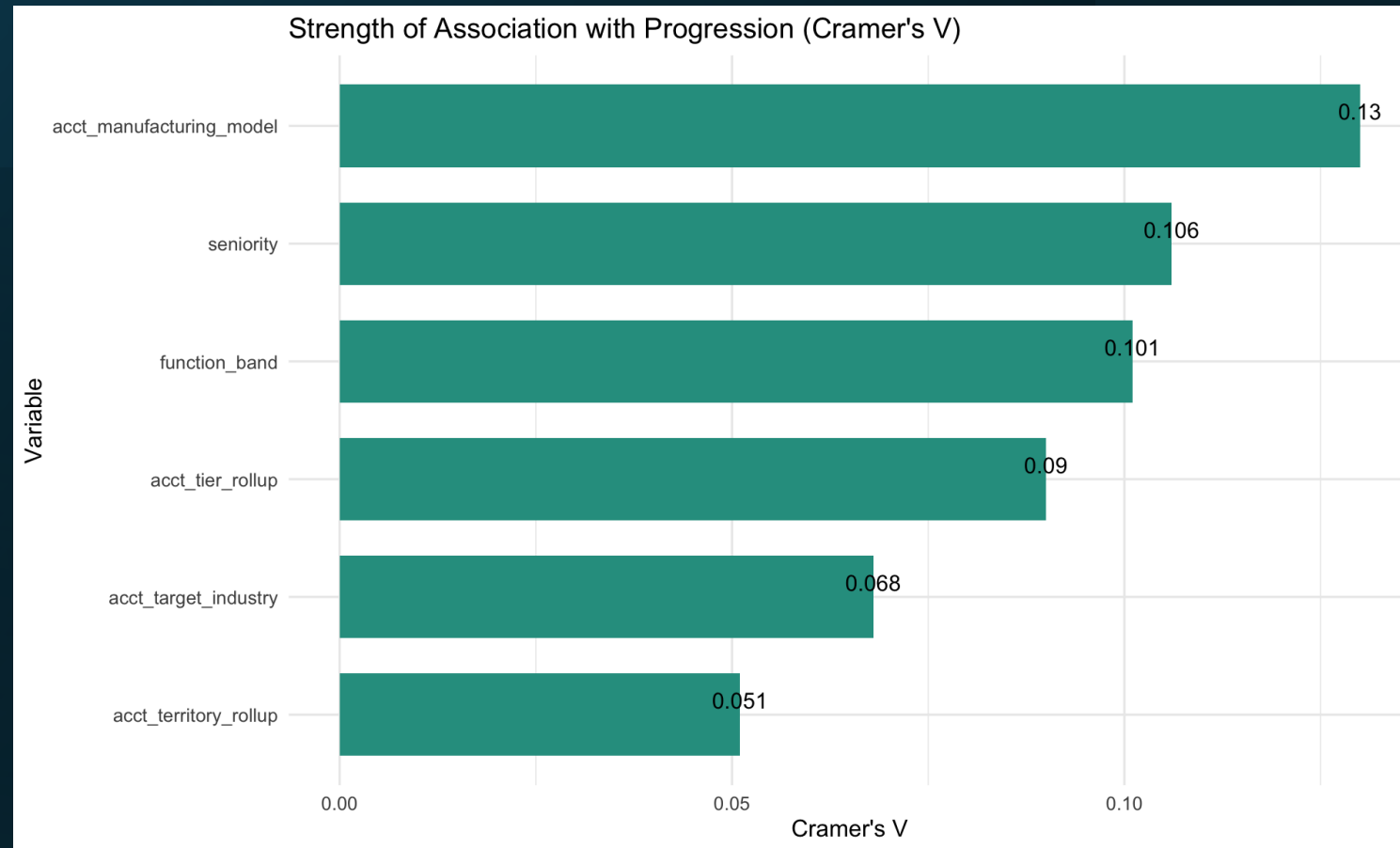
- ✓ Examined potential bias from missing variables and consolidated categorically as “unknown”

Standardized and grouped categorical variables

- ✓ standardized categorical variables (such as grouping job titles into seniority levels and functional area) to make patterns more interpretable.



Relation to Progression



Key Data Insights

Manufacturing Model

- Strongest Predictor of progression
- HOW is more important than WHAT

Buyer Persona

- Directors & Managers drive conversion, not executives

Cross Sale opportunities

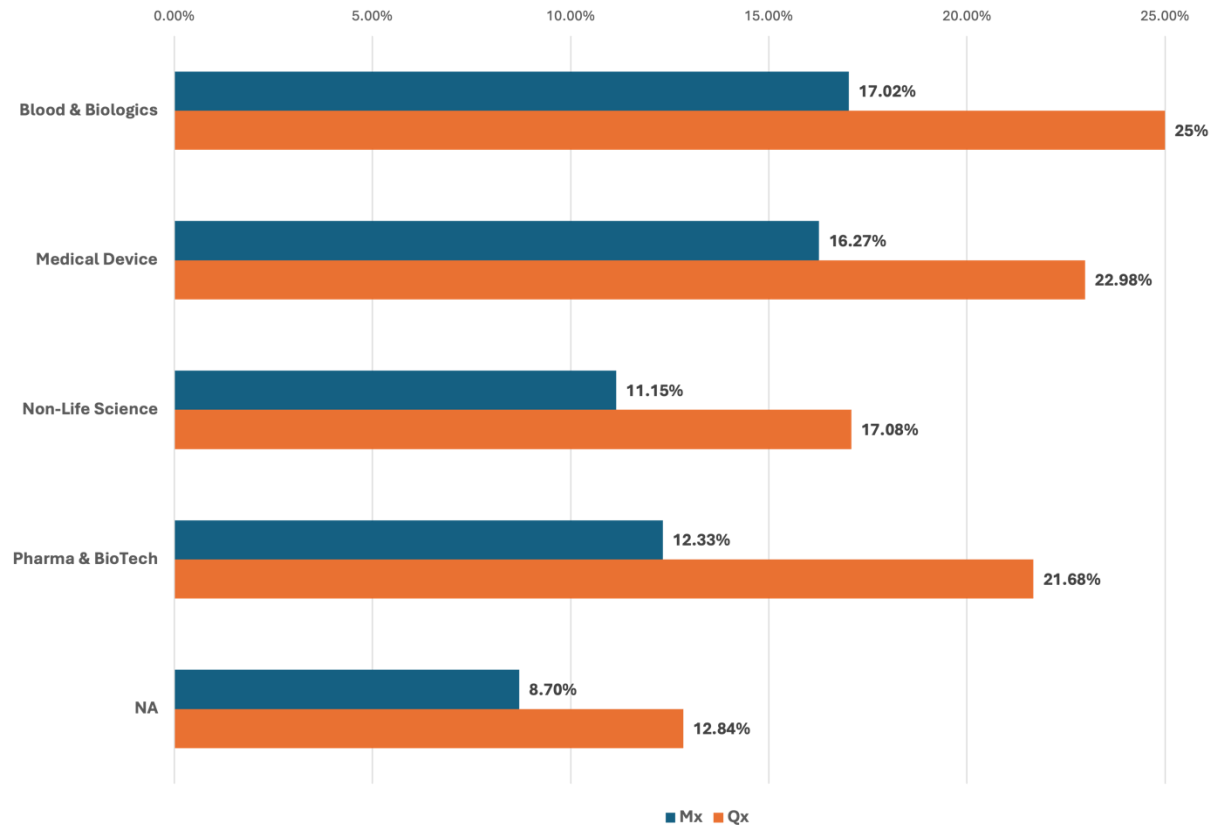
- Some profiles work for QX but not MX
- Cross sale potential for leads that have quality compliance and manufacturing roles, and managers responsible for both

Descriptive Analytics

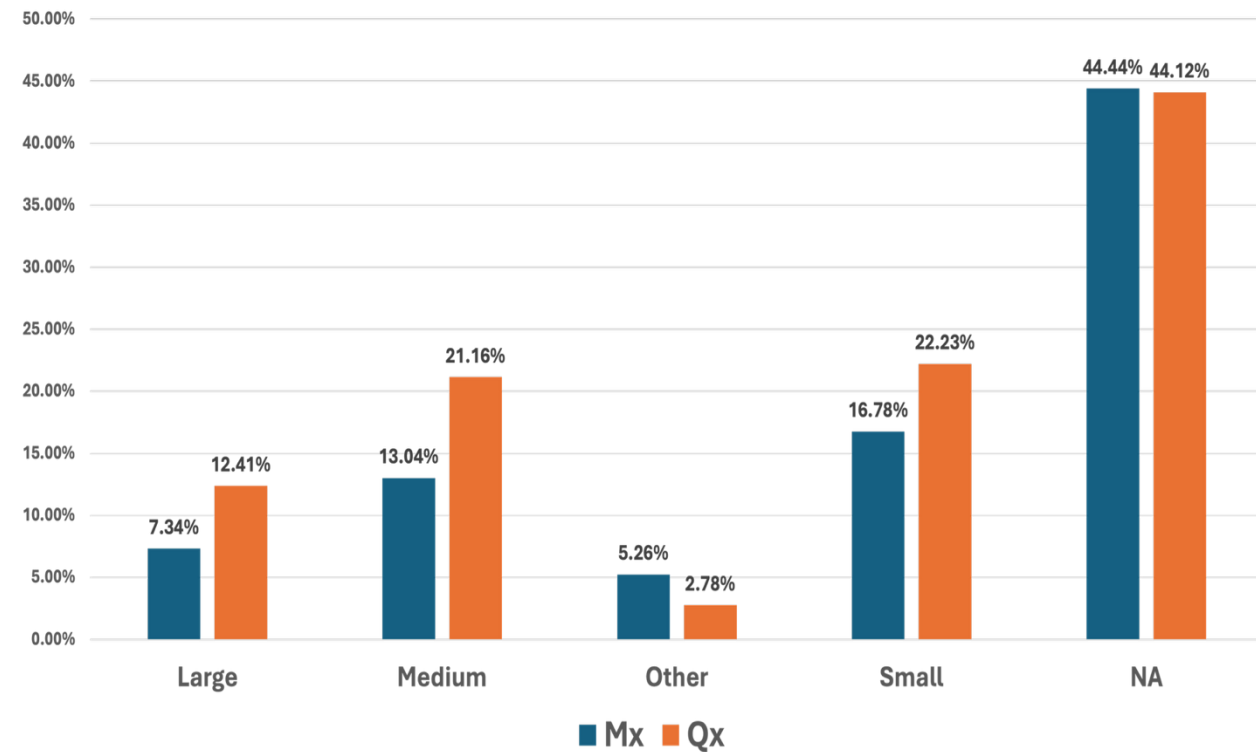
- ✓ **Raw historical data** to understand patterns
- ✓ Identify how **Mx and Qx conversion rates** vary
- ✓ Observe **natural relationships in the data** without modeling assumptions
- ✓ provide the **context for the predictive models** introduced later
- ✓ **which factors may influence progression**

Raw Data Insights

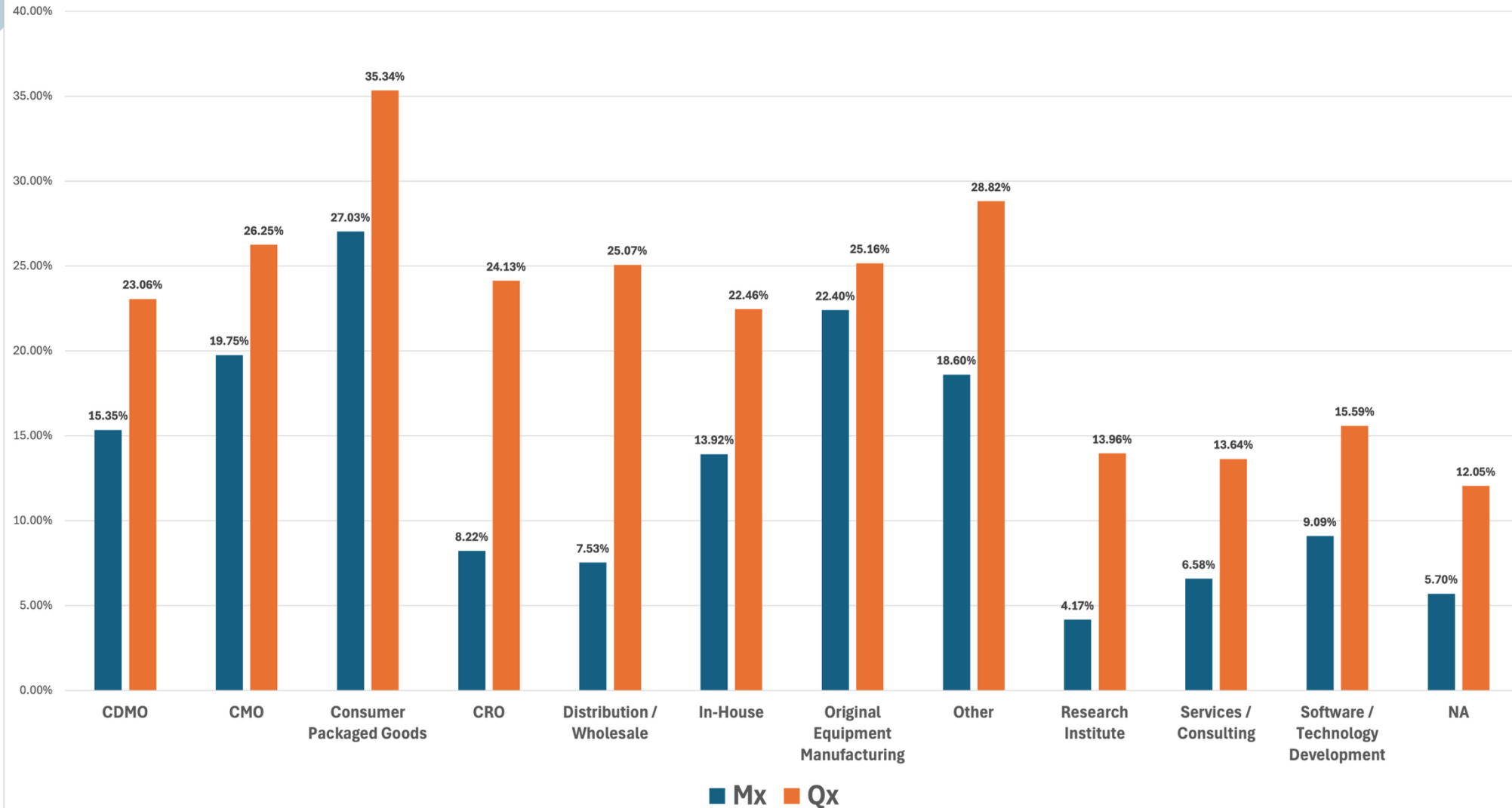
Conversion by Industry — Mx vs Qx



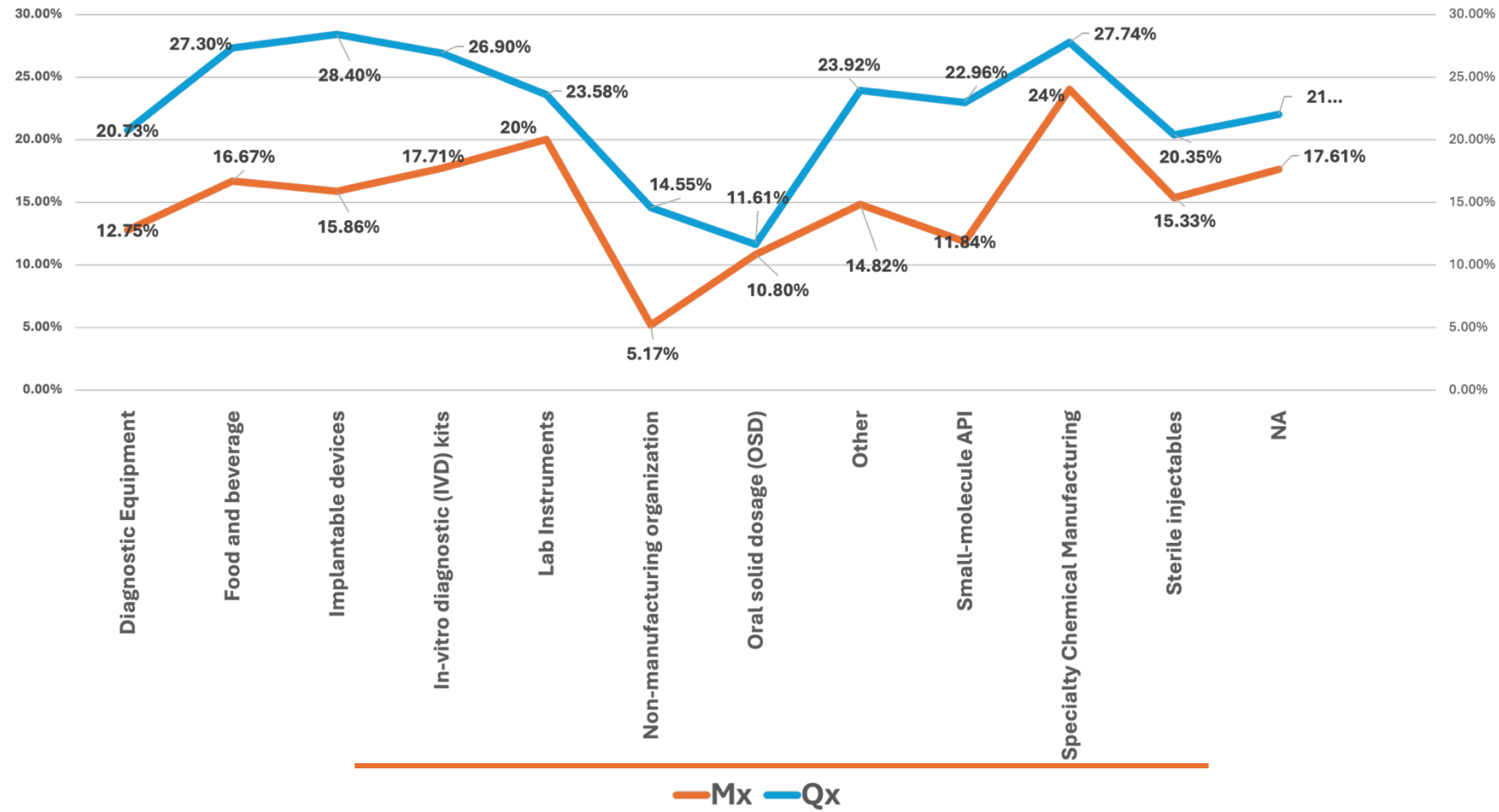
Conversion by Company Tier — Mx vs Qx



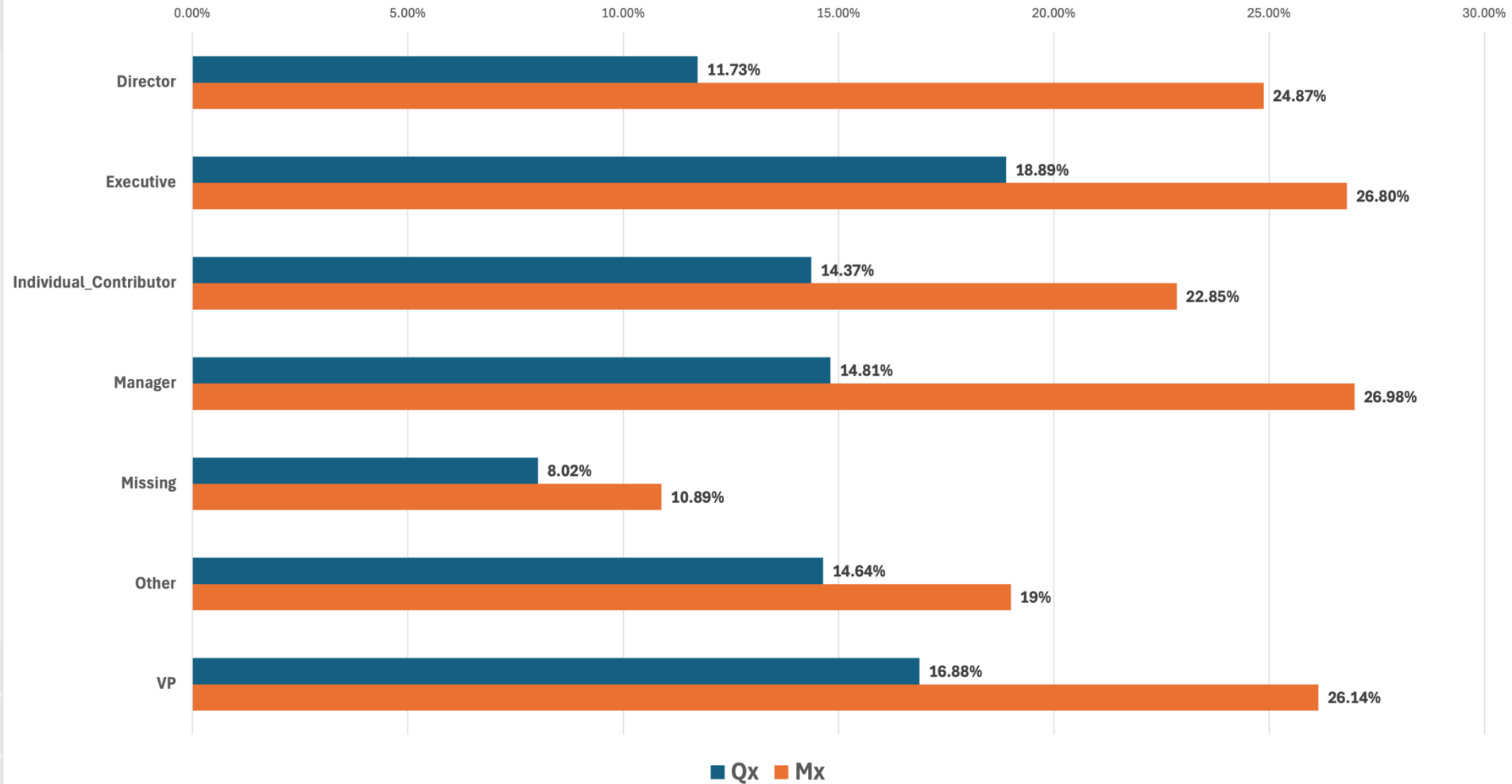
Conversion by Manufacturing Model — Mx vs Qx



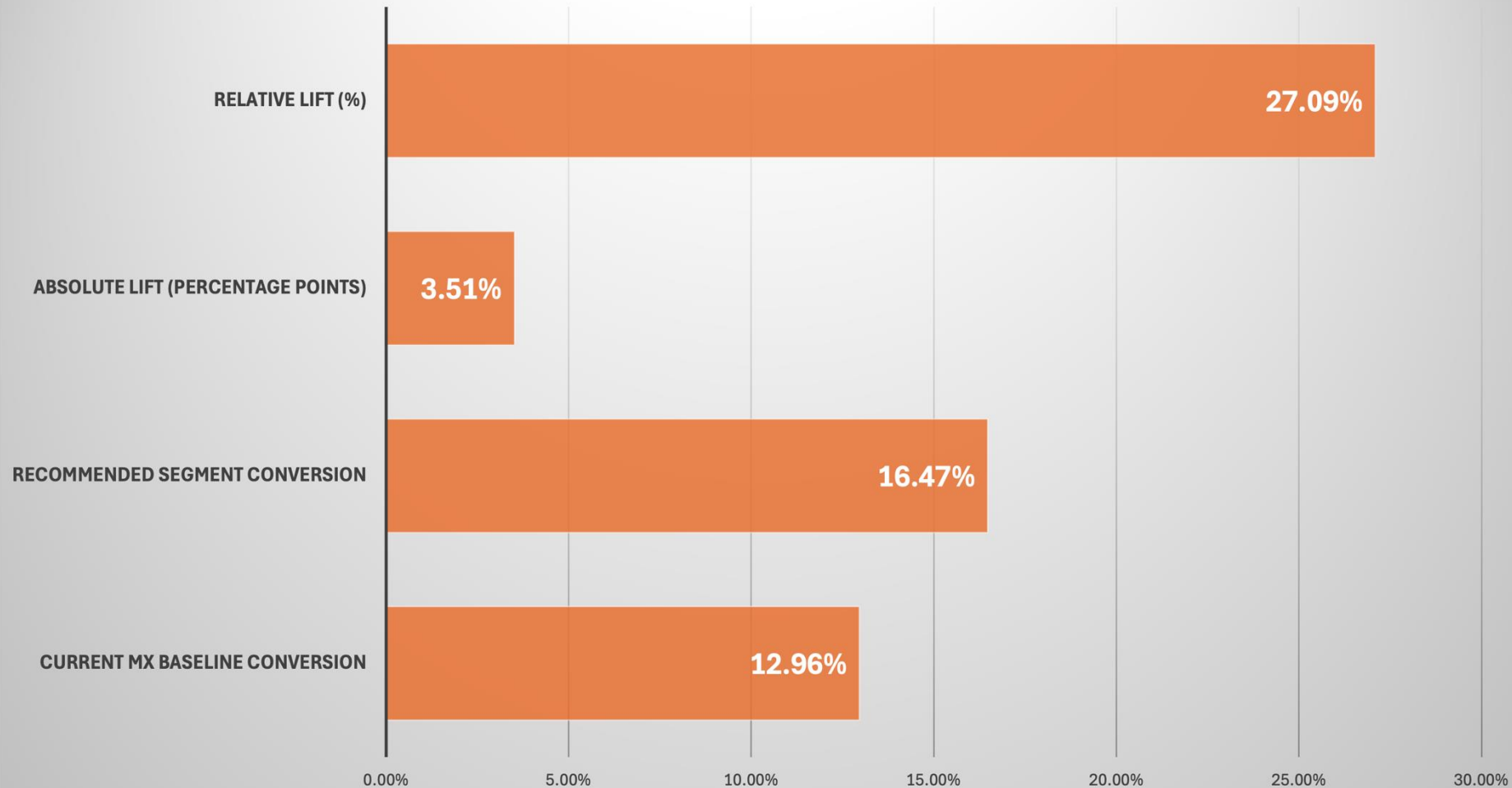
Conversion by Site Function — Mx vs Qx



Conversion by Title Seniority — Mx vs Qx

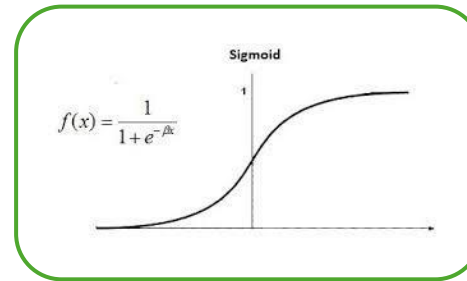


Estimated Impact of Improved Mx Targeting Value

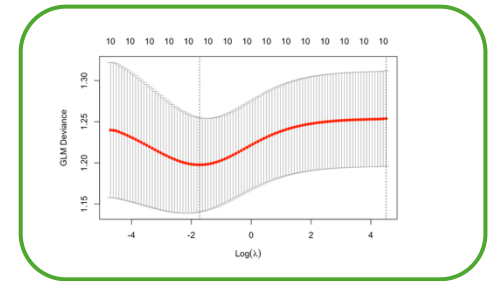


Predictive Analytics

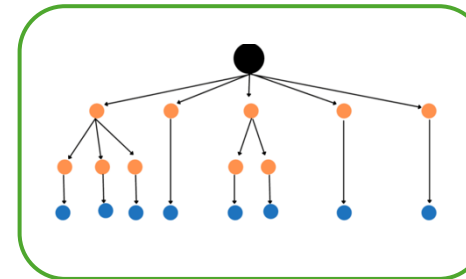
Five predictive models were evaluated:



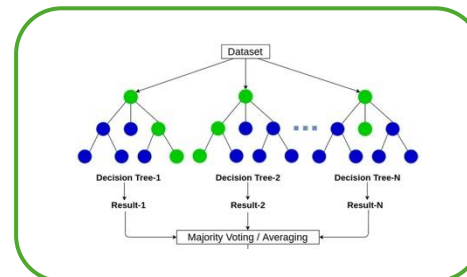
Logistic Regression Model



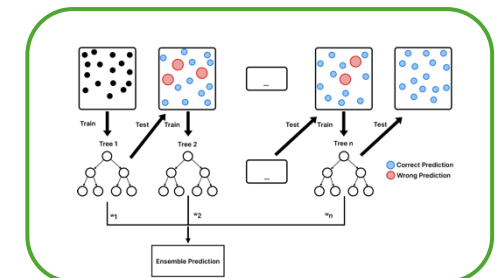
GLMNET logistic regression Model



Decision Tree Model



Random Forest Model



XGBoost Model

Why Logistic Regression?

Reason	Explanation
High interpretability	Logistic regression clearly shows how each variable (industry, company tier, role, etc.) influences the probability of progression .
Directly answers business questions	The model allows us to identify which industries, account types, and buyer roles are more likely to convert , aligning with the project objectives.
Transparent for business teams	The outputs are easy to explain to sales and marketing teams , making it easier to translate results into targeting strategies.

Understanding Model Outputs

P-value (Statistical Significance)

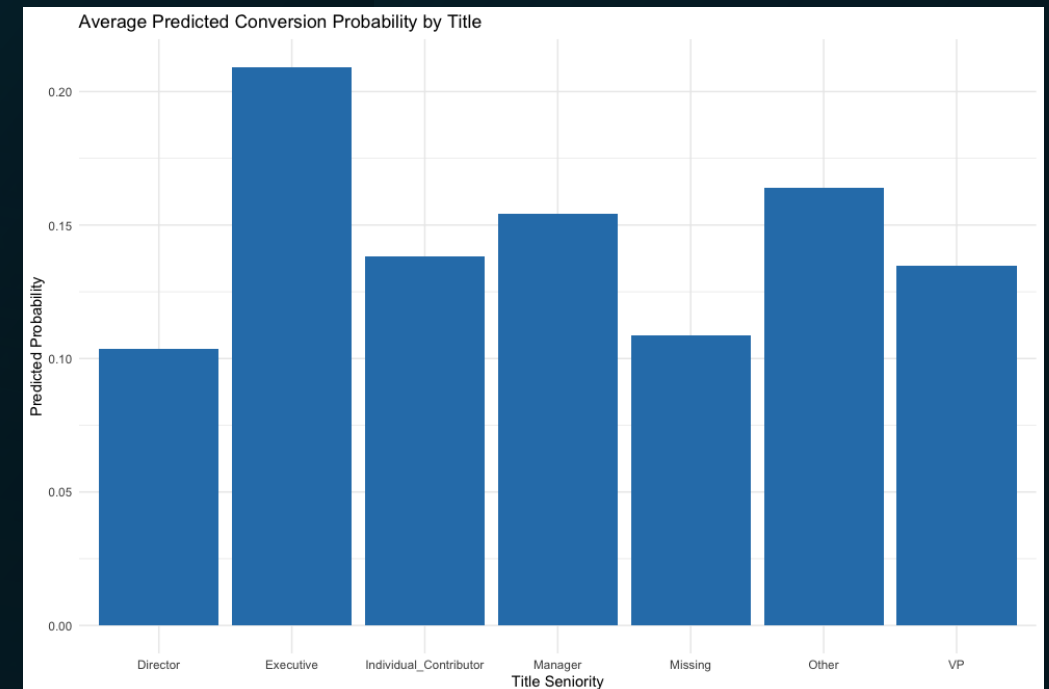
- Indicates whether a variable has a statistically meaningful relationship with progression.
- Lower p-values suggest stronger evidence that the variable influences conversion.

Odds Ratio

Shows **how much a factor increases or decreases the likelihood of progression.**

- Odds Ratio > 1 → increases likelihood of progression
- Odds Ratio < 1 → decreases likelihood of progression

Predictor	Estimate	Odds Ratio	CI Lower	CI Upper	p-value	Significance
title_seniorityManager	0.4405	1.5534	1.1210	2.1683	0.0088	** Very Significant
title_seniorityExecutive	0.6142	1.8482	1.1078	3.0391	0.0168	* Significant
title_seniorityIndividual_Contributor	0.3394	1.4042	0.9732	2.0292	0.0697	Not Significant
title_functionRegulatory	-1.6978	0.1831	0.0222	1.2180	0.0792	Not Significant
title_seniorityMissing	-0.8364	0.4333	0.1300	1.9750	0.2128	Not Significant



Prescriptive Analytics

- ✓ **Translate predictive insights into actionable strategy** for targeting and outreach.
- ✓ **Recommend where sales and marketing efforts should focus** to improve Mx conversion.
- ✓ Use model insights to **optimize targeting, messaging, and cross-sell opportunities.**
- ✓ **Goal: increase progression rates and maximize conversion impact.**

Mx Sweet Spots

High Priority

- ✓ CMO/CDMO + Small **23.9%**
- ✓ In-House + Small **19.6%**

High-Priority Segments: 32.9%

Low Priority

- ✓ Services **6.5%**
- ✓ Software **<7%**

Current Mx Baseline: 13.5%

Mx Buyer Personas

High Priority		Low Priority	
Executive	18.9%	Engineering/R&D	14.9%
VP	16.9%	Quality/Compliance	15.9%
Manager	14.8%	Software	7.8%

Current Mx Baseline: 13.5% | Best Persona: Executive or Management in Engineering

Cross-Selling

Site Function	Qx Only	Cross Sell	Mx Only
Account size	Small/Medium	Small/Medium	Small/Medium
Title	VP/Head	VP/Head	None
Industry	Medical Devices	Medical Devices	Medical Devices
Manufacturing Model	None	No-Cross sell	CMO

Actionable Recommendations

SDRs

- ✓ Target Small Manufacturers
- ✓ Focus: CMO/CDMO/In-House
- ✓ Avoid Services/Software segments

Sales Leadership

- ✓ Implement lead scoring
- ✓ Track segment performance monthly
- ✓ Validate w/ pilot program

Marketing

- ✓ Shift spend to Contract Manufacturing
- ✓ Reduce Services sector campaigns
- ✓ Focus budgets on Small tier

Expected Business Impact



310 Tier 1 Leads
Identified
(9.9% of Mx pipeline)

FROM: 13.5%
baseline
TO: 32.9% conversion

+19.4% POINTS
~2.5x improvement
potential

Next Step: Pilot →
Validate → Scale

Thank you



Questions?